Alex Gibson

github.com/AlexGibson12 · alexgibson12.github.io · Oxford, UK

EDUCATION

University of Oxford · DPhil in Statistics, EPSRC StatML CDT 2026–2030 (expected)

- Oxford-Radcliffe Scholarship
- Current research: sequential Monte Carlo methods for inference-time alignment of diffusion models

University of Cambridge · MMath and BA in Mathematics 2022–2026

- Part III: Distinction
- Part II: First Class

PUBLICATIONS

- *Compact Proofs of Model Performance via Mechanistic Interpretability.* NeurIPS 2024. Co-author.
- *Decomposing Attention to Find Context-Sensitive Neurons.* NeurIPS 2025 MechInterp Workshop. Sole author.

INDEPENDENT RESEARCH

- *Neural Networks Learn Bloom Filters* (2026). Showed that a one-layer ReLU network learns binary hash codes for tokens on a sparse top-k prediction task.
- *Ping-Pong Computation in Superposition* (2025). Exact construction for packing many small neural circuits into a single network with zero interference.

PROGRAMMING

- Python, PyTorch, C++
- Linear programming solver in C++ (simplex method, with a parser for LP input)
- Compiler for a C-like language in C++, emitting x86-64 assembly